

-----ASSIGNMENT-----

INTRO

SQL-----

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COHORT A;

-----QUESTION-----

01-----

Write the SQL code to create the table structures for the entities shown in Figure 1. The structures should contain the attributes specified in the ERD. Use data types that would be appropriate for the data that will need to be stored in each attribute. Enforce primary key and foreign key constraints as indicated by the ERD.

-----Answer;

```
MariaDB [(none)]> CREATE DATABASE dvd_kiosk;
Query OK, 1 row affected (0.005 sec)
```

```
MariaDB [(none)]> USE dvd_kiosk;
Database changed
MariaDB [dvd_kiosk]> select DATABASE();
+-----+
| DATABASE() |
+-----+
| dvd_kiosk  |
+-----+
1 row in set (0.001 sec)
```

```
MariaDB [dvd_kiosk]> CREATE TABLE MEMBERSHIP (
->     MEM_NUM INT NOT NULL,
->     MEM_FNAME VARCHAR(50) NOT NULL,
->     MEM_LNAME VARCHAR(50) NOT NULL,
->     MEM_STREET VARCHAR(100) NOT NULL,
->     MEM_CITY VARCHAR(50) NOT NULL,
->     MEM_STATE CHAR(2) NOT NULL,
->     MEM_ZIP VARCHAR(10) NOT NULL,
->     MEM_BALANCE DECIMAL(7, 2) DEFAULT 0.00,
->     CONSTRAINT PK_MEMBERSHIP PRIMARY KEY (MEM_NUM)
-> );
Query OK, 0 rows affected, 1 warning (0.004 sec)
```

```
MariaDB [dvd_kiosk]> CREATE TABLE RENTAL (
->     RENT_NUM INT NOT NULL,
->     RENT_DATE DATE NOT NULL,
->     MEM_NUM INT NOT NULL,
```

```
->     CONSTRAINT PK_RENTAL PRIMARY KEY (RENT_NUM),
->     CONSTRAINT FK_RENTAL_MEMBERSHIP FOREIGN KEY (MEM_NUM) REFERENCES
MEMBERSHIP(MEM_NUM)
-> );
```

Query OK, 0 rows affected, 1 warning (0.003 sec)

```
MariaDB [dvd_kiosk]> CREATE TABLE PRICE (
->     PRICE_CODE INT NOT NULL,
->     PRICE_DESCRIPTION VARCHAR(100) NOT NULL,
->     PRICE_RENTFEE DECIMAL(5, 2) NOT NULL,
->     PRICE_DAILYLATEFEE DECIMAL(5, 2) NOT NULL,
->     CONSTRAINT PK_PRICE PRIMARY KEY (PRICE_CODE)
-> );
```

Query OK, 0 rows affected, 1 warning (0.007 sec)

```
MariaDB [dvd_kiosk]> CREATE TABLE MOVIE (
->     MOVIE_NUM INT NOT NULL,
->     MOVIE_TITLE VARCHAR(150) NOT NULL,
->     MOVIE_YEAR INT NOT NULL,
->     MOVIE_COST DECIMAL(6, 2) NOT NULL,
->     MOVIE_GENRE VARCHAR(50) NOT NULL,
->     PRICE_CODE INT NOT NULL,
->     CONSTRAINT PK_MOVIE PRIMARY KEY (MOVIE_NUM),
->     CONSTRAINT FK_MOVIE_PRICE FOREIGN KEY (PRICE_CODE) REFERENCES
PRICE(PRICE_CODE)
-> );
```

Query OK, 0 rows affected, 1 warning (0.007 sec)

```
MariaDB [dvd_kiosk]> CREATE TABLE VIDEO (
->     VID_NUM INT NOT NULL,
->     VID_INDATE DATE NOT NULL,
->     MOVIE_NUM INT NOT NULL,
->     CONSTRAINT PK_VIDEO PRIMARY KEY (VID_NUM),
->     CONSTRAINT FK_VIDEO_MOVIE FOREIGN KEY (MOVIE_NUM) REFERENCES
MOVIE(MOVIE_NUM)
-> );
```

Query OK, 0 rows affected, 1 warning (0.007 sec)

```
MariaDB [dvd_kiosk]> CREATE TABLE DETAILRENTAL (
->     RENT_NUM INT NOT NULL,
->     VID_NUM INT NOT NULL,
->     DETAIL_FEE DECIMAL(5, 2) NOT NULL,
->     DETAIL_DUEDATE DATE NOT NULL,
->     DETAIL_RETURNDATE DATE,
->     DETAIL_DAILYLATEFEE DECIMAL(5, 2) NOT NULL,
->     CONSTRAINT PK_DETAILRENTAL PRIMARY KEY (RENT_NUM, VID_NUM),
->     CONSTRAINT FK_DETAIL_RENTAL FOREIGN KEY (RENT_NUM) REFERENCES
RENTAL(RENT_NUM),
->     CONSTRAINT FK_DETAIL_VIDEO FOREIGN KEY (VID_NUM) REFERENCES
VIDEO(VID_NUM)
```

```
-> );
Query OK, 0 rows affected, 1 warning (0.006 sec)
```

```
MariaDB [dvd_kiosk]> SHOW TABLES;
```

```
+-----+
| Tables_in_dvd_kiosk |
+-----+
| detailrental         |
| membership           |
| movie                |
| price                |
| rental               |
| video                |
+-----+
6 rows in set (0.006 sec)
```

-----QUESTION

2-----

The following tables provide a very small portion of the data that will be kept in the database. This data needs to be inserted into the database for testing purposes. Write the

INSERT commands needed to insert the following data into the created tables.

----Answer;

MEMBERSHIP TABLE

```
MariaDB [dvd_kiosk]> INSERT INTO MEMBERSHIP (MEM_NUM, MEM_FNAME, MEM_LNAME,
MEM_STREET, MEM_CITY, MEM_STATE, MEM_ZIP, MEM_BALANCE) VALUES
-> (102, 'Tami', 'Dawson', '2632 Takli Circle', 'Norene', 'TN', '37136', 11.00),
-> (103, 'Curt', 'Knight', '4025 Cornell Court', 'Flatgap', 'KY', '41219',
6.00),
-> (104, 'Jamal', 'Melendez', '788 East 145th Avenue', 'Quebeck', 'TN', '38579',
0.00),
-> (105, 'Iva', 'Mcclain', '6045 Musket Ball Circle', 'Summit', 'KY', '42783',
15.00),
-> (106, 'Miranda', 'Parks', '4469 Maxwell Place', 'Germantown', 'TN', '38183',
0.00),
-> (107, 'Rosario', 'Elliott', '757 Danner Avenue', 'Columbia', 'TN', '38402',
5.00),
-> (108, 'Mattie', 'Guy', '4390 Evergreen Street', 'Lily', 'KY', '40740', 0.00),
-> (109, 'Clint', 'Ochoa', '1711 Elm Street', 'Greeneville', 'TN', '37745',
10.00),
-> (110, 'Lewis', 'Rosales', '4524 Southwind Circle', 'Counce', 'TN', '38326',
0.00),
-> (111, 'Stacy', 'Mann', '2789 East Cook Avenue', 'Murfreesboro', 'TN',
'37132', 8.00),
-> (112, 'Luis', 'Trujillo', '7267 Melvin Avenue', 'Heiskell', 'TN', '37754',
3.00),
-> (113, 'Minnie', 'Gonzales', '6430 Vasili Drive', 'Williston', 'TN', '38076',
```

```
0.00);
Query OK, 12 rows affected (0.006 sec)
Records: 12 Duplicates: 0 Warnings: 0
```

RENTAL TABLE

```
MariaDB [dvd_kiosk]> INSERT INTO RENTAL (RENT_NUM, RENT_DATE, MEM_NUM) VALUES
-> (1001, '2016-03-01', 103),
-> (1002, '2016-03-01', 105),
-> (1003, '2016-03-02', 102),
-> (1004, '2016-03-02', 110),
-> (1005, '2016-03-02', 111),
-> (1006, '2016-03-02', 107),
-> (1007, '2016-03-02', 104),
-> (1008, '2016-03-03', 105),
-> (1009, '2016-03-03', 111);
Query OK, 9 rows affected (0.002 sec)
Records: 9 Duplicates: 0 Warnings: 0
```

PRICE TABLE

```
MariaDB [dvd_kiosk]> INSERT INTO PRICE (PRICE_CODE, PRICE_DESCRIPTION,
PRICE_RENTFEE, PRICE_DAILYLATEFEE) VALUES
-> (1, 'Standard', 2.00, 1.00),
-> (2, 'New Release', 3.50, 3.00),
-> (3, 'Discount', 1.50, 1.00),
-> (4, 'Weekly Special', 1.00, 0.50);
Query OK, 4 rows affected (0.002 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

MOVIE TABLE

```
MariaDB [dvd_kiosk]> INSERT INTO MOVIE (MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR,
MOVIE_COST, MOVIE_GENRE, PRICE_CODE) VALUES
-> (1234, 'The Cesar Family Christmas', 2014, 39.95, 'FAMILY', 2),
-> (1235, 'Smokey Mountain Wildlife', 2011, 59.95, 'ACTION', 1),
-> (1236, 'Richard Goodhope', 2015, 59.95, 'DRAMA', 2),
-> (1237, 'Beatnik Fever', 2014, 29.95, 'COMEDY', 2),
-> (1238, 'Constant Companion', 2015, 89.95, 'DRAMA', 1),
-> (1239, 'Where Hope Dies', 2005, 25.49, 'DRAMA', 3),
-> (1245, 'Time to Burn', 2012, 45.49, 'ACTION', 1),
-> (1246, 'What He Doesn''t Know', 2013, 58.29, 'COMEDY', 1);
Query OK, 8 rows affected (0.005 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

VIDEO TABLE

```
MariaDB [dvd_kiosk]> INSERT INTO VIDEO (VID_NUM, VID_INDATE, MOVIE_NUM) VALUES
-> (54321, '2015-06-18', 1234),
-> (54324, '2015-06-18', 1234),
-> (54325, '2015-06-18', 1234),
-> (34341, '2014-01-22', 1235),
-> (34342, '2014-01-22', 1235),
-> (34366, '2016-03-02', 1236),
```

```

-> (34367, '2016-03-02', 1236),
-> (34368, '2016-03-02', 1236),
-> (34369, '2016-03-02', 1236),
-> (44392, '2015-10-21', 1237),
-> (44397, '2015-10-21', 1237),
-> (59237, '2016-02-14', 1237),
-> (61388, '2014-01-25', 1239),
-> (61353, '2013-01-28', 1245),
-> (61354, '2013-01-28', 1245);

```

----- Then, I updated the video table so that it can not bring some constraint violations, because there some information where Null and the attribute did not allow it to be null.

```

MariaDB [dvd_kiosk]> INSERT INTO VIDEO (VID_NUM, VID_INDATE, MOVIE_NUM)
-> VALUES
->      (61369, '2016-03-02', 1236),
->      (61367, '2016-03-02', 1236);
Query OK, 2 rows affected (0.001 sec)
Records: 2  Duplicates: 0  Warnings: 0

```

DETAILRENTAL TABLE

```

MariaDB [dvd_kiosk]> INSERT INTO DETAILRENTAL (RENT_NUM, VID_NUM, DETAIL_FEE,
DETAIL_DUEDATE, DETAIL_RETURNDATE, DETAIL_DAILYLATEFEE) VALUES
-> (1001, 34342, 2.00, '2016-03-04', '2016-03-02', 1.00),
-> (1001, 61353, 2.00, '2016-03-04', '2016-03-03', 1.00),
-> (1002, 59237, 3.50, '2016-03-04', '2016-03-04', 3.00),
-> (1003, 54325, 3.50, '2016-03-04', '2016-03-09', 3.00),
-> -- Row 61369 omitted due to missing foreign key record in VIDEO
-> (1003, 61388, 0.00, '2016-03-06', '2016-03-09', 1.00),
-> (1004, 44392, 3.50, '2016-03-05', '2016-03-07', 3.00),
-> (1004, 34367, 3.50, '2016-03-05', '2016-03-07', 3.00),
-> (1004, 34341, 2.00, '2016-03-07', '2016-03-07', 1.00),
-> (1005, 34342, 2.00, '2016-03-07', '2016-03-05', 1.00),
-> (1005, 44397, 3.50, '2016-03-05', '2016-03-05', 3.00),
-> (1006, 34366, 3.50, '2016-03-05', '2016-03-04', 3.00),
-> -- Row 61367 omitted due to missing foreign key record in VIDEO
-> (1007, 34368, 3.50, '2016-03-05', NULL, 3.00),
-> (1008, 34369, 3.50, '2016-03-05', '2016-03-05', 3.00),
-> (1009, 54324, 3.50, '2016-03-05', NULL, 3.00),
-> (1001, 34366, 3.50, '2016-03-04', '2016-03-02', 3.00);
Query OK, 15 rows affected (0.004 sec)
Records: 15  Duplicates: 0  Warnings: 0

```

FULL TABLE;

MEMBERSHIP TABLE

```

MariaDB [dvd_kiosk]> SELECT *FROM MEMBERSHIP
-> ;

```

```

+-----+-----+-----+-----+-----+-----+-----+
---+-----+-----+

```

MEM_NUM	MEM_FNAME	MEM_LNAME	MEM_STREET	MEM_CITY	MEM_STATE	MEM_ZIP	MEM_BALANCE
102	Tami	Dawson	2632 Takli Circle	Norene	TN	37136	11.00
103	Curt	Knight	4025 Cornell Court	Flatgap	KY	41219	6.00
104	Jamal	Melendez	788 East 145th Avenue	Quebeck	TN	38579	0.00
105	Iva	Mcclain	6045 Musket Ball Circle	Summit	KY	42783	15.00
106	Miranda	Parks	4469 Maxwell Place	Germantown	TN	38183	0.00
107	Rosario	Elliott	757 Danner Avenue	Columbia	TN	38402	5.00
108	Mattie	Guy	4390 Evergreen Street	Lily	KY	40740	0.00
109	Clint	Ochoa	1711 Elm Street	Greeneville	TN	37745	10.00
110	Lewis	Rosales	4524 Southwind Circle	Counce	TN	38326	0.00
111	Stacy	Mann	2789 East Cook Avenue	Murfreesboro	TN	37132	8.00
112	Luis	Trujillo	7267 Melvin Avenue	Heiskell	TN	37754	3.00
113	Minnie	Gonzales	6430 Vasili Drive	Williston	TN	38076	0.00

12 rows in set (0.003 sec)

RENTAL TABLE

MariaDB [dvd_kiosk]> SELECT *FROM RENTAL
-> ;

RENT_NUM	RENT_DATE	MEM_NUM
1001	2016-03-01	103
1002	2016-03-01	105
1003	2016-03-02	102
1004	2016-03-02	110
1005	2016-03-02	111
1006	2016-03-02	107
1007	2016-03-02	104
1008	2016-03-03	105
1009	2016-03-03	111

9 rows in set (0.001 sec)

PRICE TABLE

MariaDB [dvd_kiosk]> SELECT *FROM PRICE

-> ;

PRICE_CODE	PRICE_DESCRIPTION	PRICE_RENTFEE	PRICE_DAILYLATEFEE
1	Standard	2.50	1.00
2	New Release	4.00	3.00
3	Discount	2.00	1.00
4	Weekly Special	1.50	0.50

4 rows in set (0.001 sec)

MariaDB [dvd_kiosk]>

MOVIE TABLE

MariaDB [dvd_kiosk]> SELECT * FROM MOVIE;

MOVIE_NUM	MOVIE_TITLE	MOVIE_YEAR	MOVIE_COST	MOVIE_GENRE
1234	The Cesar Family Christmas	2014	39.95	FAMILY
1235	Smokey Mountain Wildlife	2011	59.95	ACTION
1236	Richard Goodhope	2015	59.95	DRAMA
1237	Beatnik Fever	2014	29.95	COMEDY
1238	Constant Companion	2015	89.95	DRAMA
1239	Where Hope Dies	2005	25.49	DRAMA
1245	Time to Burn	2010	45.49	ACTION
1246	What He Doesn't Know	2013	58.29	COMEDY

8 rows in set (0.002 sec)

VIDEO TABLE

MariaDB [dvd_kiosk]> SELECT * FROM VIDEO;

VID_NUM	VID_INDATE	MOVIE_NUM
34341	2014-01-22	1235
34342	2014-01-22	1235

34366	2016-03-02	1236
34367	2016-03-02	1236
34368	2016-03-02	1236
34369	2016-03-02	1236
44392	2015-10-21	1237
44397	2015-10-21	1237
54321	2015-06-18	1234
54324	2015-06-18	1234
54325	2015-06-18	1234
59237	2016-02-14	1237
61353	2013-01-28	1245
61354	2013-01-28	1245
61367	2016-03-02	1236
61369	2016-03-02	1236
61388	2014-01-25	1239

+-----+-----+-----+

17 rows in set (0.001 sec)

DETAILRENTAL TABLE

MariaDB [dvd_kiosk]> SELECT *FROM DETAILRENTAL;

+-----+-----+-----+-----+-----+-----+

-----+

RENT_NUM	VID_NUM	DETAIL_FEE	DETAIL_DUEDATE	DETAIL_RETURNDATE	DETAIL_DAILYLATEFEE
----------	---------	------------	----------------	-------------------	---------------------

+-----+-----+-----+-----+-----+-----+

-----+

1001	34342	2.00	2016-03-04	2016-03-02	
1.00					
1001	34366	3.50	2016-03-04	2016-03-02	
3.00					
1001	61353	2.00	2016-03-04	2016-03-03	
1.00					
1002	59237	3.50	2016-03-04	2016-03-04	
3.00					
1003	54325	3.50	2016-03-04	2016-03-09	
3.00					
1003	61388	0.00	2016-03-06	2016-03-09	
1.00					
1004	34341	2.00	2016-03-07	2016-03-07	
1.00					
1004	34367	3.50	2016-03-05	2016-03-07	
3.00					
1004	44392	3.50	2016-03-05	2016-03-07	
3.00					
1005	34342	2.00	2016-03-07	2016-03-05	
1.00					
1005	44397	3.50	2016-03-05	2016-03-05	
3.00					
1006	34366	3.50	2016-03-05	2016-03-04	
3.00					

1007	34368	3.50	2016-03-05	NULL
3.00				
1008	34369	3.50	2016-03-05	2016-03-05
3.00				
1009	54324	3.50	2016-03-05	NULL
3.00				

-----+
-----+
15 rows in set (0.001 sec)

-----QUESTION
3-----

Write the SQL command to change the movie year for movie number 1245 to 2010. Show the result before and after the change.

-----Answer,
----before
MariaDB [dvd_kiosk]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE
-> WHERE MOVIE_NUM = 1245;
+-----+
| MOVIE_NUM | MOVIE_TITLE | MOVIE_YEAR |
+-----+
| 1245 | Time to Burn | 2012 |
+-----+
1 row in set (0.003 sec)

----after
MariaDB [dvd_kiosk]> UPDATE MOVIE
-> SET MOVIE_YEAR =2010
-> WHERE MOVIE_NUM =1245;
Query OK, 1 row affected (0.003 sec)
---- the graph
MariaDB [dvd_kiosk]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR
-> FROM MOVIE
-> WHERE MOVIE_NUM = 1245;
+-----+
| MOVIE_NUM | MOVIE_TITLE | MOVIE_YEAR |
+-----+
| 1245 | Time to Burn | 2010 |
+-----+
1 row in set (0.001 sec)

-----QUESTI
ON
4-----
Write the SQL command to change the price code for all Action movies to price code 3.Show

the result before and after the change.

---Answer,

---before

```
MariaDB [dvd_kiosk]> SELECT MOVIE_NUM,MOVIE_TITLE, MOVIE_GENRE, PRICE_CODE
-> FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_GENRE	PRICE_CODE
1235	Smokey Mountain Wildlife	ACTION	1
1245	Time to Burn	ACTION	1

---After,

```
MariaDB [dvd_kiosk]> UPDATE MOVIE
```

```
-> SET PRICE_CODE = 3
```

```
-> WHERE MOVIE_GENRE = 'ACTION';
```

Query OK, 2 rows affected (0.005 sec)

Rows matched: 2 Changed: 2 Warnings: 0

MOVIE_NUM	MOVIE_TITLE	MOVIE_GENRE	PRICE_CODE
1235	Smokey Mountain Wildlife	ACTION	3
1245	Time to Burn	ACTION	3

-----QUESTIO

N 5-----

Write a single SQL command to increase all price rental fee values in the PRICE table by

\$0.50. Show the result before and after the change.

-----Answer,

-----before

```
MariaDB [dvd_kiosk]> SELECT PRICE_CODE, PRICE_DESCRIPTION, PRICE_RENTFEE
-> FROM PRICE;
```

PRICE_CODE	PRICE_DESCRIPTION	PRICE_RENTFEE
1	Standard	2.00
2	New Release	3.50
3	Discount	1.50
4	Weekly Special	1.00

-----after

```
MariaDB [dvd_kiosk]> UPDATE PRICE
```

```
-> SET PRICE_RENTFEE = PRICE_RENTFEE + 0.50;
```

Query OK, 4 rows affected (0.002 sec)
Rows matched: 4 Changed: 4 Warnings: 0

PRICE_CODE	PRICE_DESCRIPTION	PRICE_RENTFEE
1	Standard	2.50
2	New Release	4.00
3	Discount	2.00
4	Weekly Special	1.50

-----QUESTION

6-----

Write a query to display the movie title, movie year, and movie genre for all movies (result shown in Figure 2).

---Answer,

MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE
-> FROM MOVIE;

MOVIE_TITLE	MOVIE_YEAR	MOVIE_GENRE
The Cesar Family Christmas	2014	FAMILY
Smokey Mountain Wildlife	2011	ACTION
Richard Goodhope	2015	DRAMA
Beatnik Fever	2014	COMEDY
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
Time to Burn	2010	ACTION
What He Doesn't Know	2013	COMEDY

8 rows in set (0.002 sec)

-----QUESTION

7-----

Write a query to display the movie year, movie title, and movie cost sorted by movie year in descending order (result shown in Figure 3).

---Answers

MariaDB [dvd_kiosk]> SELECT MOVIE_YEAR, MOVIE_TITLE, MOVIE_COST
-> FROM MOVIE
-> ORDER BY MOVIE_YEAR DESC;

MOVIE_YEAR	MOVIE_TITLE	MOVIE_COST
2015	Richard Goodhope	59.95

2015	Constant Companion	89.95
2014	The Cesar Family Christmas	39.95
2014	Beatnik Fever	29.95
2013	What He Doesn't Know	58.29
2011	Smokey Mountain Wildlife	59.95
2010	Time to Burn	45.49
2005	Where Hope Dies	25.49

+-----+
8 rows in set (0.068 sec)

-----QUESTI
ON

8-----
Write a query to display the movie title, movie year, and movie genre for all movies sorted by movie genre in ascending order, then sorted by movie year in descending order within genre (result shown in Figure 4).

-----Answers

MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE
-> FROM MOVIE
-> ORDER BY MOVIE_GENRE ASC, MOVIE_YEAR DESC;

MOVIE_TITLE	MOVIE_YEAR	MOVIE_GENRE
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION
Beatnik Fever	2014	COMEDY
What He Doesn't Know	2013	COMEDY
Richard Goodhope	2015	DRAMA
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
The Cesar Family Christmas	2014	FAMILY

-----QUESTION
9-----

-
Write a query to display the movie number, movie title, and price code for all movies with a title that starts with the letter "R" (result shown in Figure 5).

----Answer

MariaDB [dvd_kiosk]> SELECT MOVIE_NUM, MOVIE_TITLE, PRICE_CODE
-> FROM MOVIE
-> WHERE MOVIE_TITLE LIKE 'R%';

MOVIE_NUM	MOVIE_TITLE	PRICE_CODE
-----------	-------------	------------

1236	Richard Goodhope	2
------	------------------	---

-----QUESTION
N

10-----
Write a query to display the movie title, movie year, and movie cost for all movies that contain the word "hope" anywhere in the title. Sort the results in ascending order by title (result shown in Figure 6).

-----Answer,
MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_COST
-> FROM MOVIE
-> WHERE MOVIE_TITLE LIKE '%HOPE%'
-> ORDER BY MOVIE_TITLE ASC;

MOVIE_TITLE	MOVIE_YEAR	MOVIE_COST
Richard Goodhope	2015	59.95
Where Hope Dies	2005	25.49

2 rows in set (0.002 sec)

-----QUESTION
11-----

Write a query to display the movie title, movie year, and movie genre for all action movies (result shown in Figure 7)

-----Answer,
MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE
-> FROM MOVIE
-> WHERE MOVIE_GENRE = 'ACTION';

MOVIE_TITLE	MOVIE_YEAR	MOVIE_GENRE
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION

2 rows in set (0.002 sec)

-----QUESTION
12-----

--
Write a query to display the movie number, movie title, and movie cost for all movies with a

cost greater than \$40 (result shown in Figure 8).

-----Answer

```
MariaDB [dvd_kiosk]> SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_COST
-> FROM MOVIE
-> WHERE MOVIE_COST > 40.00;
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_COST
1235	Smokey Mountain Wildlife	59.95
1236	Richard Goodhope	59.95
1238	Constant Companion	89.95
1245	Time to Burn	45.49
1246	What He Doesn't Know	58.29

-----QUEST
ION

13-----
Write a query to display the movie number, movie title, movie cost, and movie genre for all movies that are either action or comedy movies and that have a cost that is less than \$50. Sort the results in ascending order by genre. (Result shown in Figure 9.)

----Answer,

```
MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_COST, MOVIE_GENRE
-> FROM MOVIE
-> WHERE (MOVIE_GENRE = 'ACTION' OR MOVIE_GENRE = 'COMEDY')
-> AND MOVIE_COST < 50.00
-> ORDER BY MOVIE_GENRE ASC;
```

MOVIE_TITLE	MOVIE_COST	MOVIE_GENRE
Time to Burn	45.49	ACTION
Beatnik Fever	29.95	COMEDY

2 rows in set (0.001 sec)

-----QUES
TION

14-----
Write a query to display the membership number, name, street, state, and balance for all members in Tennessee (TN), with a balance less than \$5, and whose street name ends in "Avenue." (result shown in Figure 10).

---Answer,

```
MariaDB [dvd_kiosk]> SELECT MEM_NUM, MEM_FNAME, MEM_LNAME, MEM_STREET, MEM_STATE,
```

MEM_BALANCE

```
-> FROM MEMBERSHIP
-> WHERE MEM_STATE = 'TN'
-> AND MEM_BALANCE < 5.00
-> AND MEM_STREET LIKE '%Avenue';
```

MEM_NUM	MEM_FNAME	MEM_LNAME	MEM_STREET	MEM_STATE	MEM_BALANCE
104	Jamal	Melendez	788 East 145th Avenue	TN	0.00
112	Luis	Trujillo	7267 Melvin Avenue	TN	3.00

-----QUEST
ION

15-----
Write a query to display the movie genre and the number of movies in each genre (result shown in Figure 11).

----Answer,
MariaDB [dvd_kiosk]> SELECT MOVIE_GENRE, COUNT(*) AS "Number of Movies"
-> FROM MOVIE
-> GROUP BY MOVIE_GENRE;

MOVIE_GENRE	Number of Movies
ACTION	2
COMEDY	2
DRAMA	3
FAMILY	1

4 rows in set (0.006 sec)

-----QUESTIO
N

16-----
Write a query to display the average cost of all of the movies (result shown in Figure 12).

---Answer,
MariaDB [dvd_kiosk]> SELECT AVG(MOVIE_COST)
-> FROM MOVIE;

AVG(MOVIE_COST)

51.127500

-----QUESTI
ON

17-----
Write a query to display the movie genre and average cost of movies in each genre (result shown in Figure P7.13).

-----Answer,
MariaDB [dvd_kiosk]> SELECT MOVIE_GENRE, AVG(MOVIE_COST) AS "Average Cost"
-> FROM MOVIE
-> GROUP BY MOVIE_GENRE;

MOVIE_GENRE	Average Cost
ACTION	52.720000
COMEDY	44.120000
DRAMA	58.463333
FAMILY	39.950000

-----QUESTION
18-----
-

Write a query to display the movie title, movie genre, price description, and price rental fee for all movies with a price code (result shown in Figure 14).

-----Answer,
MariaDB [dvd_kiosk]> SELECT M.MOVIE_TITLE, M.MOVIE_GENRE, P.PRICE_DESCRIPTION, P.PRICE_RENTFEE
-> FROM MOVIE M
-> INNER JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE
-> WHERE M.PRICE_CODE IS NOT NULL;

MOVIE_TITLE	MOVIE_GENRE	PRICE_DESCRIPTION	PRICE_RENTFEE
The Cesar Family Christmas	FAMILY	New Release	4.00
Smokey Mountain Wildlife	ACTION	Discount	2.00
Richard Goodhope	DRAMA	New Release	4.00
Beatnik Fever	COMEDY	New Release	4.00
Constant Companion	DRAMA	Standard	2.50
Where Hope Dies	DRAMA	Discount	2.00
Time to Burn	ACTION	Discount	2.00
What He Doesn't Know	COMEDY	Standard	2.50

-----QUESTIO

N

19-----

Write a query to display the movie genre and average price rental fee for movies in each genre that have a price (result shown in Figure 15).

----Answer,

MariaDB [dvd_kiosk]> SELECT M.MOVIE_GENRE, AVG(P.PRICE_RENTFEE) AS "Average Rental Fee"

-> FROM MOVIE M

-> INNER JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE

-> WHERE M.PRICE_CODE IS NOT NULL

-> GROUP BY M.MOVIE_GENRE;

MOVIE_GENRE	Average Rental Fee
ACTION	2.000000
COMEDY	3.250000
DRAMA	2.833333
FAMILY	4.000000

-----QUESTI

ON

20-----

Write a query to display the movie title and the movie cost divided by the price rental fee for each movie that has a price to determine the number of rentals it will take to break even on the purchase of the movie (result shown in Figure 16).

----Answer,

MariaDB [dvd_kiosk]> SELECT M.MOVIE_TITLE, (M.MOVIE_COST / P.PRICE_RENTFEE) AS "Breakeven Rentals"

-> FROM MOVIE M

-> INNER JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE

-> WHERE M.PRICE_CODE IS NOT NULL;

MOVIE_TITLE	Breakeven Rentals
The Cesar Family Christmas	9.987500
Smokey Mountain Wildlife	29.975000
Richard Goodhope	14.987500
Beatnik Fever	7.487500
Constant Companion	35.980000
Where Hope Dies	12.745000
Time to Burn	22.745000
What He Doesn't Know	23.316000

-----QUESTI
ON

21-----
. Write a query to display the movie title and movie year for all movies that have a price code (result shown in Figure 17).

---Answer,
MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_YEAR
-> FROM MOVIE
-> WHERE PRICE_CODE IS NOT NULL;

MOVIE_TITLE	MOVIE_YEAR
The Cesar Family Christmas	2014
Smokey Mountain Wildlife	2011
Richard Goodhope	2015
Beatnik Fever	2014
Constant Companion	2015
Where Hope Dies	2005
Time to Burn	2010
What He Doesn't Know	2013

-----QUESTIO
N

22-----
Write a query to display the movie title, movie genre, and movie cost for all movies that have a cost between \$44.99 and \$49.99 (result shown in Figure 18)

---Answer,
MariaDB [dvd_kiosk]> SELECT MOVIE_TITLE, MOVIE_GENRE, MOVIE_COST
-> FROM MOVIE
-> WHERE MOVIE_COST BETWEEN 44.99 AND 49.99;

MOVIE_TITLE	MOVIE_GENRE	MOVIE_COST
Time to Burn	ACTION	45.49

-----QUESTION
23-----

-
Write a query to display the movie title, price description, and price rental fee for all movies that are in the genres Family, Comedy, or Drama (result shown in Figure 19).

--Answer,

```

MariaDB [dvd_kiosk]> SELECT M.MOVIE_TITLE, P.PRICE_DESCRIPTION, P.PRICE_RENTFEE
-> FROM MOVIE M
-> INNER JOIN PRICE P ON M.PRICE_CODE = P.PRICE_CODE
-> WHERE M.MOVIE_GENRE IN ('FAMILY', 'COMEDY', 'DRAMA');

```

MOVIE_TITLE	PRICE_DESCRIPTION	PRICE_RENTFEE
The Cesar Family Christmas	New Release	4.00
Richard Goodhope	New Release	4.00
Beatnik Fever	New Release	4.00
Constant Companion	Standard	2.50
Where Hope Dies	Discount	2.00
What He Doesn't Know	Standard	2.50

-----QUEST
ION

24-----
Write a query to display the membership number, first name, last name, and balance of the memberships that have a rental (result shown in Figure 20).

---Answer,

```

MariaDB [dvd_kiosk]> SELECT DISTINCT M.MEM_NUM, M.MEM_FNAME, M.MEM_LNAME,
M.MEM_BALANCE
-> FROM MEMBERSHIP M
-> INNER JOIN RENTAL R ON M.MEM_NUM = R.MEM_NUM;

```

MEM_NUM	MEM_FNAME	MEM_LNAME	MEM_BALANCE
102	Tami	Dawson	11.00
103	Curt	Knight	6.00
104	Jamal	Melendez	0.00
105	Iva	Mcclain	15.00
107	Rosario	Elliott	5.00
110	Lewis	Rosales	0.00
111	Stacy	Mann	8.00

-----QUESTI
ON

25-----
Write a query to display the minimum balance, maximum balance, and average balance for memberships that have a rental (result shown in Figure 21).

---Answer,

```

MariaDB [dvd_kiosk]> SELECT MIN(M.MEM_BALANCE) AS "Minimum Balance",
-> MAX(M.MEM_BALANCE) AS "Maximum Balance",

```

```

->      AVG(M.MEM_BALANCE) AS "Average Balance"
-> FROM MEMBERSHIP M
-> INNER JOIN RENTAL R ON M.MEM_NUM = R.MEM_NUM;

```

Minimum Balance	Maximum Balance	Average Balance
0.00	15.00	7.555556

-----QUESTI
ON

26-----

Write a query to display the rental number, rental date, video number, movie title, due date, and return date for all videos that were returned after the due date. Sort the results by rental number and movie title (result shown in Figure 22).

---Answer,

```

MariaDB [dvd_kiosk]> SELECT R.RENT_NUM,
->      R.RENT_DATE,
->      D.VID_NUM,
->      M.MOVIE_TITLE,
->      D.DETAIL_DUEDATE,
->      D.DETAIL_RETURNDATE
-> FROM RENTAL R
-> INNER JOIN DETAILRENTAL D ON R.RENT_NUM = D.RENT_NUM
-> INNER JOIN VIDEO V      ON D.VID_NUM = V.VID_NUM
-> INNER JOIN MOVIE M      ON V.MOVIE_NUM = M.MOVIE_NUM
-> WHERE D.DETAIL_RETURNDATE > D.DETAIL_DUEDATE
-> ORDER BY R.RENT_NUM ASC, M.MOVIE_TITLE ASC;

```

RENT_NUM	RENT_DATE	VID_NUM	MOVIE_TITLE	DETAIL_DUEDATE	DETAIL_RETURNDATE
1003	2016-03-02	54325	The Cesar Family Christmas	2016-03-04	2016-03-09
1003	2016-03-02	61388	Where Hope Dies	2016-03-06	2016-03-09
1004	2016-03-02	44392	Beatnik Fever	2016-03-05	2016-03-07
1004	2016-03-02	34367	Richard Goodhope	2016-03-05	2016-03-07

QUESTION

27-----

. Write a query to display the rental number, rental date, movie title, and detail fee for each movie that was returned on or before the due date (result shown in Figure 23).

---Answer,

MariaDB [dvd_kiosk]> SELECT R.RENT_NUM,

-> R.RENT_DATE,

-> M.MOVIE_TITLE,

-> D.DETAIL_FEE

-> FROM RENTAL R

-> INNER JOIN DETAILRENTAL D ON R.RENT_NUM = D.RENT_NUM

-> INNER JOIN VIDEO V ON D.VID_NUM = V.VID_NUM

-> INNER JOIN MOVIE M ON V.MOVIE_NUM = M.MOVIE_NUM

-> WHERE D.DETAIL_RETURNDATE <= D.DETAIL_DUEDATE;

RENT_NUM	RENT_DATE	MOVIE_TITLE	DETAIL_FEE
1001	2016-03-01	Smokey Mountain Wildlife	2.00
1001	2016-03-01	Richard Goodhope	3.50
1001	2016-03-01	Time to Burn	2.00
1002	2016-03-01	Beatnik Fever	3.50
1004	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Beatnik Fever	3.50
1006	2016-03-02	Richard Goodhope	3.50
1008	2016-03-03	Richard Goodhope	3.50

QUESTION

28-----

Write a query to display the movie number, movie genre, average movie cost of movies in that

genre, movie cost of that individual movie, and the percentage difference between the average

movie cost and the individual movie cost (result shown in Figure 24). Note: the percentage

difference is calculated as the cost of the individual movie minus the average cost of movies in

that genre, divided by the average cost of movies in that genre multiplied by 100.

For example,

if the average cost of movies in the "Family" genre is \$25, if a given Family movie cost \$26, then

the calculation would be $((26 - 25) / 25 * 100)$, which would work out to be 4.00%.

This

indicates that this movie costs 4% more than the average Family movie.

---Answer,

MariaDB [dvd_kiosk]> SELECT M.MOVIE_NUM,

-> M.MOVIE_GENRE,

-> ROUND(AVG_TABLE.AvgCost, 2) AS "Average Cost",

-> M.MOVIE_COST,

-> ROUND(((M.MOVIE_COST - AVG_TABLE.AvgCost) / AVG_TABLE.AvgCost * 100),

2) AS "Percent Difference"

-> FROM MOVIE M

-> INNER JOIN (

-> SELECT MOVIE_GENRE, AVG(MOVIE_COST) AS AvgCost

-> FROM MOVIE

-> GROUP BY MOVIE_GENRE

->) AVG_TABLE ON M.MOVIE_GENRE = AVG_TABLE.MOVIE_GENRE;

MOVIE_NUM	MOVIE_GENRE	Average Cost	MOVIE_COST	Percent Difference
1234	FAMILY	39.95	39.95	0.00
1235	ACTION	52.72	59.95	13.71
1236	DRAMA	58.46	59.95	2.54
1237	COMEDY	44.12	29.95	-32.12
1238	DRAMA	58.46	89.95	53.86
1239	DRAMA	58.46	25.49	-56.40
1245	ACTION	52.72	45.49	-13.71
1246	COMEDY	44.12	58.29	32.12

8 rows in set (0.014 sec)